UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Calculate -86 - (-68) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

QUESTION A2 **(5 marks)**

a) List TWO (2) advantages of SMP organization.

b) Define cluster organization.

c) Give ONE (1) benefit of cluster organization.

QUESTION A3 **(5 marks)**

The advancement of technology through different computer generations has focused on improving performance, speed, and efficiency. From vacuum tubes in the first generation to modern AI-powered systems, each generation introduced faster processing, smaller components, and greater reliability.

a) Name the inventor of the following devices:

   i. Mechanical Calculating Machine

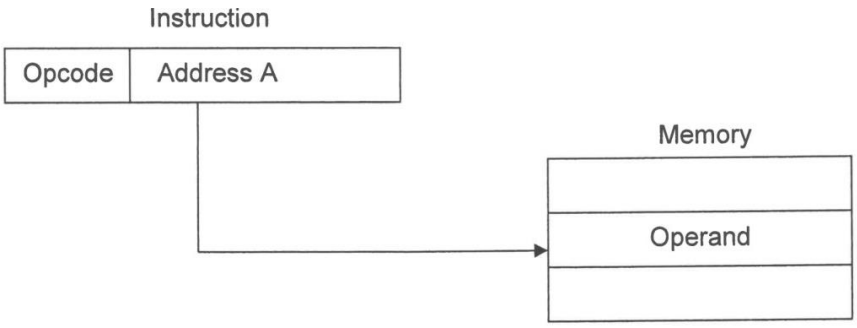
   ii. Analytical Machine iiD IAS Machine

b) Why are the machines of later computer generation designed for general purposes instead of the eadier computer generation that is more specific in function?

QUESTION A4 **(5 marks)**

Given the following Figure 2.

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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A5 **(10 marks)**

a) Compare system interconnection and CPU interconnection in terms of:

i. function

ii. physical appearance.

b) Describe the function of the THREE (3) different bus types in a computer system.

QUESTION A6 **(10 marks)**

a) List FOUR (4) types of access method for storage device with ONE (1) example for each of them.

b) Calculate the total of memory location using 11 address bit.

c) Calculate the address bits required for 2MB of RAM.

QUESTION A7 **(10 marks)**

a) Define interrupt.

b) Differentiate between interrupt and direct manipulation access.

c) Given the following scenario. Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed

messages.

Are you sure you want to There is a hardware event failure in this computer. Please reboot cancel this request? this computer now.

(a)

(b)

i. Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.

ii. While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

QUESTION A8 **(10 marks)**

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of keyboard characters.

b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

QUESTION A9 **(10 marks)**

a) Differentiate between organization and architecture.

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

QUESTION A10 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.

Expression: 50 - (-64)

a) Calculate 50 - (-64) using sign and magnitude.

b) Calculate 50 - (-64) using 1's and 2's complement.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		